SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

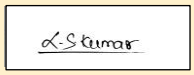
ASSESSMENT TOOL 1 – CONCEPT MAP

Course Code	**ITA0333**	**Course Title**	**Mobile computing**
CO Assessed	**CO1**	**Assessment**	**Assessment Tool 1 – Concept Map**
Weightage	**60 % of CIA**	**Total Marks**	**20 marks**
CO1 Statement	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
Student Name	**L. Sanjay Kumar**	**Reg. No.**	**192425226**
Section/Batch	**Slot D / 2024**	**Date**	**28/10/2026**

Code of Conduct:

*I, **L. Sanjay Kumar (192425226)**, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



Instructions

- This test contains ONE question.
- Duration: 30 minutes (At home)

Rubrics for Calculation:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	4	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	6	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solvi ng Approach	4	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	4	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	2	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

MOBILE COMPUTING

ASSESSMENT TOOL 1 – CONCEPT MAP

Topic: Evaluation of Mobile Roaming Paradigms, Agreements, and Settlement Architectures

Question 18: Evaluation of Roaming Types, Inter-Operator Agreements, and Settlement Challenges

Task: Develop a concept map that evaluates different types of roaming.

- ***Compare intra-PLMN, national and international roaming, including roaming agreements and inter-operator settlement.***
- ***Assess the technical and commercial challenges in each type.***
- ***Support your evaluation using logical concept organisation.***

Executive Overview: In cellular telecommunication architectures, roaming is the core operational capability that guarantees seamless communication continuity for mobile subscribers venturing beyond the geographic coverage footprint of their registered Home Public Land Mobile Network (HPLMN). While traditional cellular mobility models focus on intra-cell and inter-cell radio handoffs within a localized Base Station Subsystem (BSS), roaming governs mobility across administrative, geographical, and sovereign network boundaries. Evaluating roaming requires a rigorous multi-tier conceptual model spanning signalling plane synchronization, voice and packet routing optimizations (call tromboning and local breakout), standardized contractual frameworks governed by the GSMA (such as AA.12/AA.13 bilateral agreements), and complex financial clearinghouse operations mediated by Transferred Account Procedures (TAP3 / BCE). This assessment provides a comprehensive concept-mapped evaluation of Intra-PLMN, National, and International roaming paradigms, analyzing their underlying technical mechanisms, commercial settlement models, and operational challenges.

1. Master Concept Map: Multi-Tier Roaming Taxonomy

The foundational architecture of mobile roaming is organized into three distinct tiers dictated by the administrative domain, Mobile Country Code (MCC), Mobile Network Code (MNC), and physical boundary crossed:

- **Intra-PLMN Roaming:** Mobility occurring entirely within a single operator's network (identical MCC+MNC) across internal Visited Location Register (VLR), Mobile Switching Centre (MSC), or Access and Mobility Management Function (AMF) service areas. Signalling is handled strictly by internal Mobile Application Part (MAP-D) or Diameter S6a interfaces with zero inter-operator financial friction.

- **National Roaming:** Mobility between two distinct domestic mobile network operators sharing the same geographic country (identical MCC, differing MNC). Governed by domestic bilateral commercial contracts and national regulatory mandates, it is deployed primarily to bridge rural coverage blackspots or during Multi-Operator Core Network (MOCN) infrastructure sharing.
- **International Roaming:** Cross-border mobility across independent sovereign operators in different countries (differing MCC and MNC). Requires standardized GSMA bilateral agreements, international IP Packet eXchange (IPX) transport carrier transit, complex multi-currency TAP3 financial clearing, and wholesale inter-operator tariff (IOT) settlements.

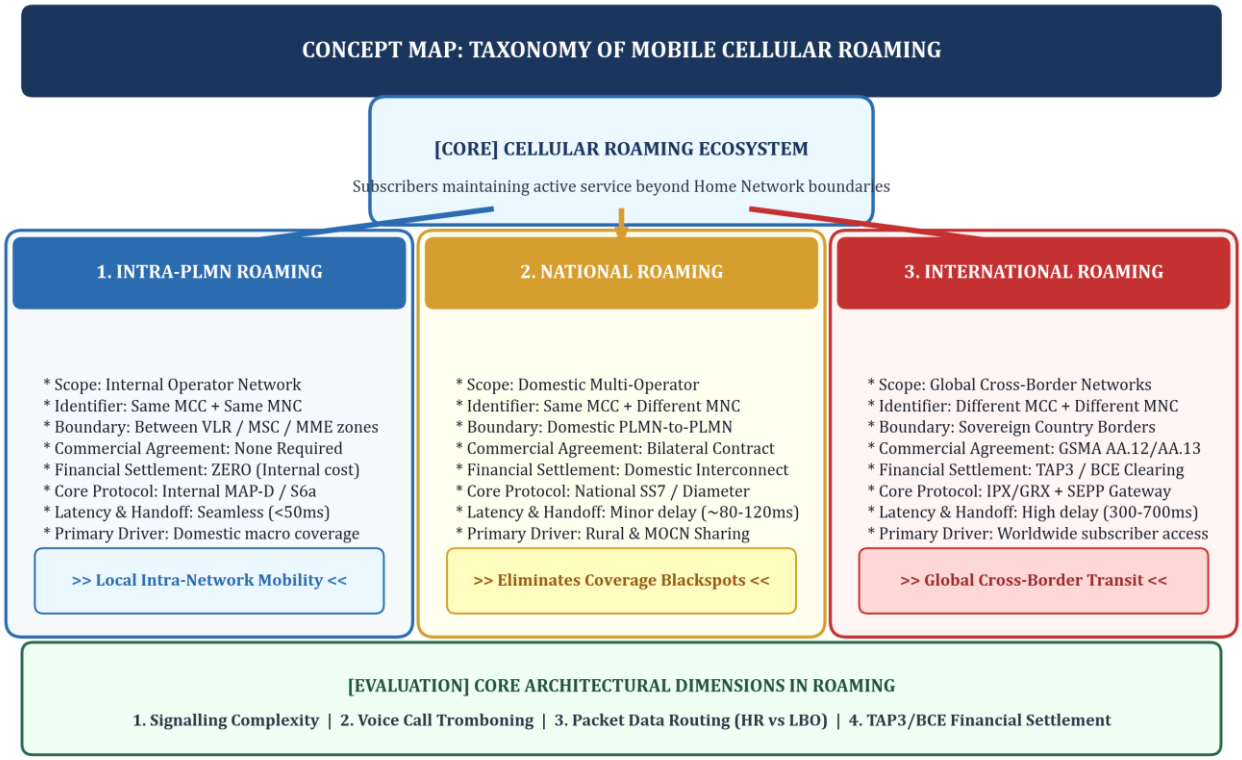


Figure 1: Comprehensive Architectural Taxonomy of Mobile Roaming Paradigms and Core Evaluation Dimensions.

2. Comparative Architectural Matrix: Intra-PLMN, National & International Roaming

To systematically compare the operational realities of each roaming type, the following structured comparative matrix analyzes network boundaries, protocol stacks, commercial agreements, and clearing mechanisms:

Architectural Dimension	Intra-PLMN Roaming	National Roaming	International Roaming
Network Domain	Single Operator (Same MCC, Same MNC)	Different Operators, Same Country (Same MCC, Diff MNC)	Independent Operators, Different Countries (Diff MCC & MNC)
Geographic Scope	Internal coverage zones within national footprint	Domestic territory (across rural/urban licensed zones)	Cross-border sovereign nations and international transit
Commercial Agreement	None required (Operator-internal provisioning)	Domestic Bilateral Wholesale Agreement (often regulated)	GSMA Standard Bilateral Agreements (GSMA PRD AA.12 / AA.13)
Signalling Protocol	Internal MAP (SS7) / Diameter S6a / 5G SBI (N8/N12)	National Interconnect SS7 / Diameter Interconnect	International IPX / GRX transit via Diameter Routing Agents (DRA)
Voice Call Delivery	Direct internal routing via local Visited MSC	National Gateway MSC (GMSC) trunk routing	International GMSC-to-VMSC routing (Prone to Tromboning)
Data Packet Routing	Direct Local P-GW / UPF breakout	Home Routed or Local Breakout via National Peering	Home Routed (S8HR) or Local Breakout (LBO) via IPX
Billing & Clearing	Internal CDRs via Operator Billing System	Domestic TAP/RAP or national wholesale CDR billing	GSMA TAP3 (TD.57) / BCE (TD.201) through Data Clearinghouse
Currency & Exchange Risk	Single local domestic currency (Zero Forex risk)	Single national currency (Zero Forex risk)	Special Drawing Rights (SDR), USD, or EUR (High Forex risk)
Regulatory Oversight	Standard national spectrum licensing terms	National Telecom Regulatory Authority (TRAI, FCC, Ofcom)	ITU-T Regulations, Regional Wholesale Caps (EU RLAH)

3. Signalling Architecture: Location Update, Authentication & Call Routing

The technical execution of roaming relies upon continuous location registration and authentication synchronization between the Visited Network (VPLMN) and the Home Network (HPLMN):

- **Phase 1: Radio Attachment & IMSI Acquisition:** Upon entering a foreign network, the Mobile Station (MS) scans broadcast channels (BCCH) and identifies a supported PLMN from

its SIM Preferred PLMN List. It issues a Location Update Request across the Um radio interface to the Visited MSC/VLR containing its International Mobile Subscriber Identity (IMSI).

- **Phase 2: International Location Update & HLR Interrogation:** Recognizing the subscriber belongs to a foreign network (via the MCC/MNC prefix of the IMSI), the Visited VLR transmits a MAP_UPDATE_LOCATION message across the international IPX/SS7 backbone to the subscriber's Home HLR/AuC (or sends a 3GPP Diameter ULR to the Home HSS/UDM).
- **Phase 3: Cryptographic Authentication & Profile Download:** The Home AuC generates authentication triplets/quintuplets (RAND, SRES, Kc in GSM; RAND, AUTN, XRES, CK, IK in UMTS/LTE) and dispatches them via MAP_INSERT_SUBSCRIBER_DATA to the Visited VLR. The VLR challenges the MS, authenticates the cryptographic response, assigns a localized Temporary Mobile Subscriber Identity (TMSI), and authorizes network access.
- **Phase 4: Mobile-Terminated Call Delivery & The Tromboning Problem:** When a caller dials the roaming subscriber's public directory number (MSISDN), the call routes first to the Home Gateway MSC (GMSC). The GMSC interrogates the Home HLR, which contacts the Visited VLR to obtain a temporary Mobile Station Roaming Number (MSRN). The GMSC then forwards the physical voice trunk to the Visited MSC using the MSRN. In international roaming, this creates the notorious Call Tromboning phenomenon—where a call from a local caller in the visited country must travel internationally to the home country and all the way back, accumulating multi-hundred-millisecond propagation latency and double international transit toll charges.

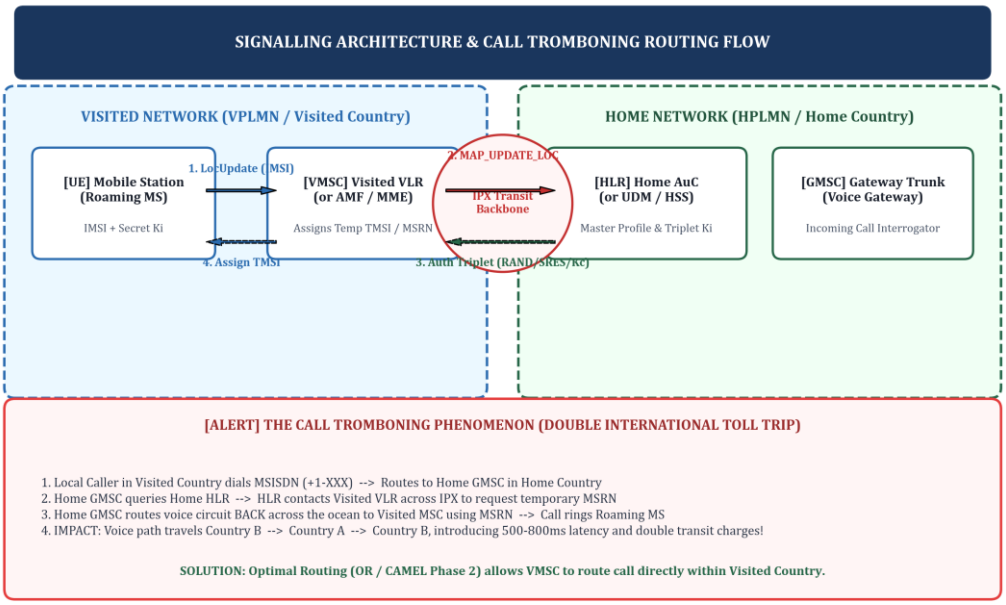


Figure 2: End-to-End Roaming Signalling Flow: Registration, Authentication Vectors, and Call Tromboning Routing.

4. Packet Data Roaming: Home Routed (S8HR) vs. Local Breakout (LBO)

With the dominance of 4G LTE and 5G NR data traffic, mobile operators must balance user Quality of Experience (QoE) against commercial billing control. Two architectural models govern packet routing during roaming:

Technical Parameter	Home Routed Roaming (S8HR / HR)	Local Breakout Roaming (LBO)	Architectural Trade-off Assessment
Packet Egress Point	Home Packet Data Network Gateway (H-PGW / H-UPF)	Visited Packet Data Network Gateway (V-PGW / V-UPF)	HR centralizes traffic in home country; LBO egresses at foreign boundary.
IPX Transit Path	Entire user payload encapsulated across GTP-U tunnels	User payload never traverses international IPX link	HR creates massive international bandwidth consumption on IPX links.
Latency & Round-Trip Time	High latency (200ms – 600ms due to intercontinental hop)	Ultra-low latency (10ms – 40ms local routing)	LBO is dramatically superior for real-time gaming, VoIP, and video calls.
Billing & Charging Visibility	Home Online Charging System (OCS) retains real-time control	Requires complex Diameter Gy/Gz synchronization with VPLMN	HR guarantees 100% fraud prevention and bill-shock quota enforcement.
Value-Added Services	Home content filters, corporate APNs, and parental controls apply	Bypasses home security filters; subject to visited regulations	HR ensures consistent corporate policy enforcement across foreign borders.

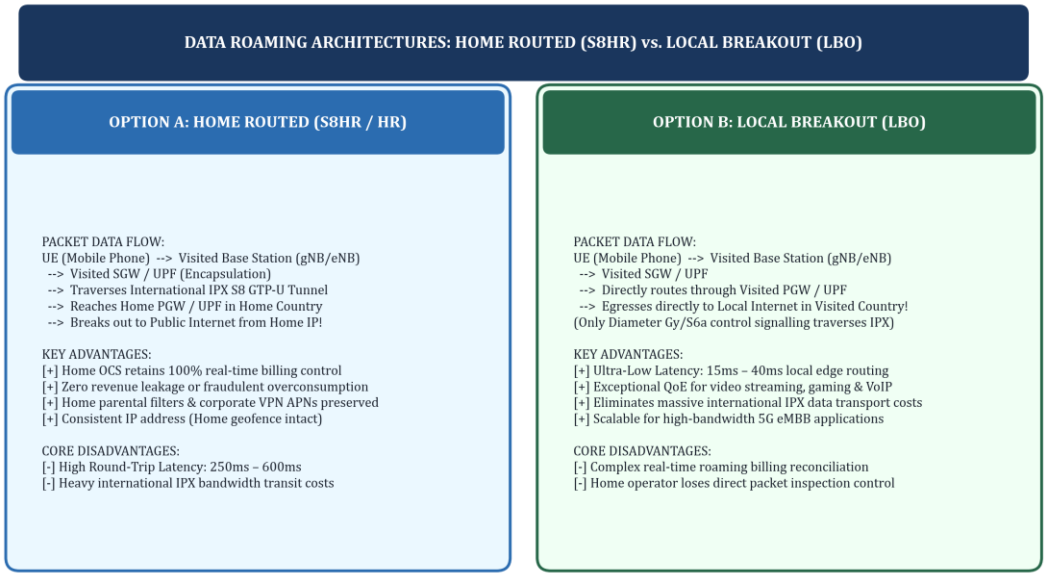


Figure 3: Data Roaming Architecture Comparison: Home Routed (S8HR) vs. Local Breakout (LBO).

5. Roaming Agreements, Clearinghouse Settlement & Financial Lifecycle

Behind the technical signalling interfaces lies an intricate global commercial and financial infrastructure that enables competing network operators to exchange traffic and settle multi-million-dollar billing accounts deterministically:

5.1 Standardized GSMA Roaming Contract Framework

- **GSMA PRD AA.12 / AA.13:** The international standard bilateral agreement template. It legally governs technical parameters, supported frequency bands, QoS commitments, fault escalation matrices, and wholesale pricing structures.
- **Inter-Operator Tariffs (IOT) & Discount Agreements:** Wholesale pricing schedules negotiated between operators. Operators establish bilateral volume commitments—pledging specific traffic volumes in exchange for discounted wholesale rates per gigabyte or voice minute.

5.2 Financial Clearing Workflow: From CDR to Net Settlement

- **1. Call Detail Record (CDR) Generation:** The Visited MSC and Serving Gateway record every call, SMS, and data session initiated by the inbound roaming subscriber.
- **2. Transferred Account Procedure (TAP3) File Creation:** Under GSMA PRD TD.57, the VPLMN converts raw CDRs into standardized ASN.1-encoded TAP3 records containing IMSI, timestamps, volume, and wholesale SDR charges, and dispatches them within 24-48 hours to a certified Data Clearinghouse (DCH).
- **3. Data Clearinghouse (DCH) Validation & RAP Protocol:** The DCH (e.g., Syniverse, Comfone, TOMIA) validates TAP3 files against formal business rules. Corrupted or invalid records are rejected via Returned Accounts Procedure (RAP) files (GSMA TD.35) for re-billing.
- **4. Financial Clearinghouse (FCH) Netting & Multi-Currency Settlement:** The FCH consolidates debts between bilateral operator pairs across multiple international currencies, applies agreed-upon IOT discounts, calculates a single monthly net financial balance in Special Drawing Rights (SDR) or USD, and facilitates bank wire transfers.
- **5. Billing and Charging Evolution (BCE - GSMA PRD TD.201):** Designed for 5G, massive IoT, and network slicing, BCE replaces batch TAP3 files with flexible, automated wholesale reconciliation, handling billions of low-power IoT devices efficiently.

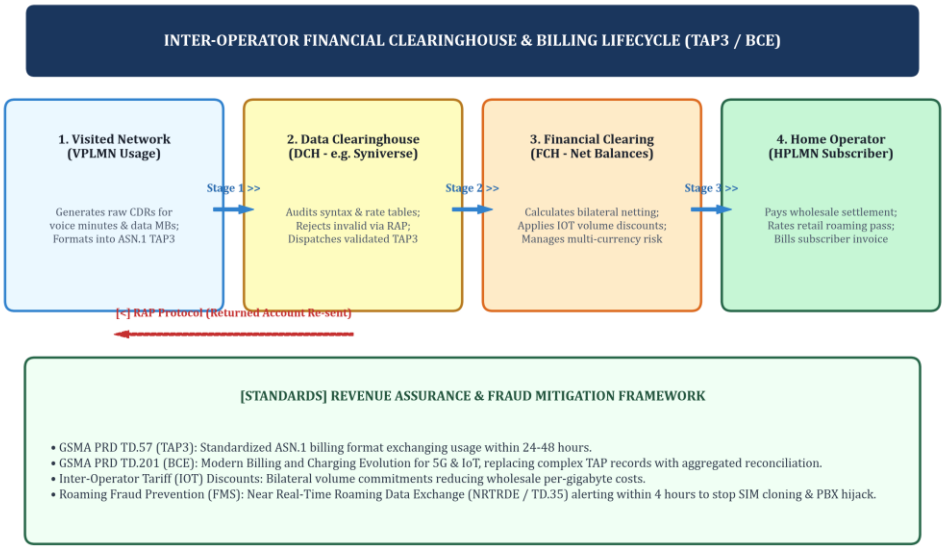


Figure 4: Inter-Operator Financial Settlement & Data Clearinghouse Workflow (TAP3 / BCE Lifecycle).

6. Comprehensive Evaluation: Technical & Commercial Challenges

A rigorous evaluation of roaming models reveals distinct operational friction points across technical engineering and commercial economics:

Roaming Paradigm	Core Technical Challenges	Core Commercial & Financial Challenges	Mitigation Strategy & Modern Architecture
Intra-PLMN Roaming	• High signalling load during massive peak-hour commuter migrations • VLR memory database synchronization limits • Intersystem handoff latency (2G to 4G/5G transitions)	• Purely internal capital and operational expense • Zero external wholesale revenue generation • Internal budget allocation across regional network circles	• Dynamic cloud-native VLR/AMF scaling in NFV core • Tracking Area List (TAL) optimization to minimize location update storms.
National Roaming	• RF boundary interference between adjacent spectrum operators • Disparate Core Network vendor interoperability (Ericsson vs. Nokia) • Core QoS degradation when VPLMN prioritizes its own native users	• Asymmetric traffic balances (rural operators become net debtors) • Competitor cannibalization (disincentivizes rural infrastructure rollout) • Complex regulatory caps enforced by national telecom bodies	• Multi-Operator Core Network (MOCN) and MORAN architecture sharing • Strict SLA and QoS enforcement for inbound guest subscribers • Regulated cost-plus wholesale pricing models
International Roaming	• Call Tromboning latency (voice round-trip lag >600ms) • Incompatible global RF spectrum bands (e.g., US 700MHz vs. EU 800MHz) • S8HR VoLTE IMS interoperability and emergency call routing failures • High packet round-trip time across intercontinental transit IPX	• Severe Foreign Exchange (Forex) currency fluctuation risk • Inter-Operator Tariff (IOT) billing disputes and RAP file rejections • Complex regulatory mandates (EU Roam-Like-At-Home eliminating margins) • Incurring unrecoverable SIM cloning before detection	• Deployment of Diameter Edge Agents (DEA) and SEPP for 5G security • Adoption of GSMA BCE (Billing and Charging Evolution) automated settlement • Local Breakout (LBO) data peering and synthetic monitoring probes

7. Concept Relationship Synthesis & Analytical Conclusion

Logical Concept Linkages: The relationship governing roaming effectiveness can be synthesized through the following core technical and commercial dependencies:

- **Administrative Scope dictates Signaling Complexity:** Intra-PLMN requires only localized MAP/Diameter database lookups; crossing national and sovereign borders mandates complex multi-hop IPX transit and DRA routing.
- **Routing Architecture determines Latency & User Experience:** Home Routed (S8HR) maintains perfect operator billing control at the expense of propagation latency, while Local Breakout (LBO) delivers optimal speed but increases billing reconciliation overhead.
- **Standardized Contracts enable Global Financial Interoperability:** GSMA AA.12/AA.13 bilateral agreements and automated TAP3/BCE data clearinghouses are essential to eliminate financial settlement disputes across fluctuating global currency markets.

Conclusion: Seamless roaming is the crowning architectural achievement of cellular engineering, transforming isolated regional networks into a unified global communications fabric. While Intra-PLMN and National roaming successfully resolve internal and domestic coverage boundaries, international roaming demands a sophisticated convergence of SS7/Diameter signalling, secure IPX transit backbones, standardized GSMA bilateral treaties, and automated financial clearinghouses. As 5G Standalone (5G SA) and massive IoT deployments scale globally, the transition from batch TAP3 clearing to real-time Billing and Charging Evolution (BCE) combined with Security Edge Protection Proxies (SEPP) will ensure resilient, low-latency, and fraud-resistant international roaming for decades to come.